Yiyang Huang

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Ph.D. candidate at Northeastern University focusing on tool-using, self-improving multimodal agents, video understanding, and visual hallucination, with research internships at Adobe Research and first-author publications at ICLR, ECCV, ACL, and EMNLP.

Education

Northeastern University — Boston, MA
Ph.D. in Computer Engineering, advised by Prof. Yun Raymond Fu — *Sep 2024 – May 2029 (Expected)*

Xidian University — Xi'an, China
M.S. in Computer Science, advised by Prof. Xuefeng Liang — *Sep 2021 – Jun 2024*
B.Eng. in Intelligence Science and Technology — *Sep 2017 – Jun 2021*

Selected First-Author Publications [Full List]

[1] **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Jianglin Lu, Qihua Dong, Hailing Wang, Huimin Zeng, Mingyuan Zhang, Yun Fu. "RACER: Reflective Agent Coupling Query Interpretation and Tool-Based Retrieval for Frame Selection in Long Video Understanding." ***Preprint***. **[Paper]**

[2] **Yiyang Huang**, Liang Shi, Yitian Zhang, Yi Xu, Yun Fu. "SHIELD: Suppressing Hallucinations in LVLM Encoders via Bias and Vulnerability Defense." **ICLR 2026**. **[Paper] [Code]**

[3] **Yiyang Huang**, Zhaowen Wang, Simon Jenni, Jing Shi, Yitian Zhang, Yizhou Wang, Yun Fu. "Beyond Atomic Layouts: Compositional Design Understanding with Vision-Language Models." **ECCV 2026**. **[Paper] [Code] [Project Page]**

[4] **Yiyang Huang**, Yitian Zhang, Yizhou Wang, Mingyuan Zhang, Liang Shi, Huimin Zeng, Yun Fu. "Distorted or Fabricated? A Survey on Hallucination in Video LLMs." **ACL 2026** (Findings). **[Paper] [Repo]**

[5] **Yiyang Huang**, Yizhou Wang, Yun Fu. "D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression and Question Decomposition." **EMNLP 2025** (Main). **[Paper] [Code]**

Research Experience

Adobe Research — San Jose, CA
Research Intern | *Mentors: Dr. Zhaowen Wang, Dr. Shraman Pramanick* — May 2026 – Nov 2026 (Expected)

[Ongoing] Learning to Critique and Refine with a Self-Improving VLM Evaluator

- Constructed the DesignJudge dataset and post-trained Qwen3.5-9B and Qwen3.8-27B for structured visual critique and scoring.
- Developed a critique–edit–score loop pairing a VLM evaluator with a Codex editor, using self-generated preference pairs for iterative GRPO updates of the evaluator.
- The post-trained evaluator achieved 72.4% mean vote-weighted agreement with human designers' judgments across Crello and internal Adobe product data, compared with Gemini-3.5-Flash (69.5%), Claude-Opus-5 (67.3%), and GPT-5.6-Sol (55.8%).

Northeastern University — Boston, MA
Ph.D. student | Supervisor: Prof. Yun Raymond Fu — Sep 2024 – Present

[*Preprint*] Reflective Agent with Query Decomposition and Tool-Based Retrieval for Long-Video Understanding

- Developed RACER, a training-free reflective agent combining query decomposition, tool-based frame retrieval, and query refinement from visual feedback. Improved accuracy by 4.3–6.6 percentage points over uniform sampling across Video-MME, MLVU, and LongVideoBench with four Vid-LLMs, with gains up to 11.6 points on MLVU under a 32-frame answering budget.

[ICLR'26] Mitigating Visual Hallucinations in MLLMs

- Identified encoder-side causes of hallucination and developed SHIELD, combining inference-time token interventions with adversarial contrastive decoding to achieve a 27% relative reduction in captioning hallucinations across multiple MLLM families.

[EMNLP'25] Training-Free Adaptation of Image-Pretrained MLLMs for Video Understanding

- Developed D-CoDe to adapt image-pretrained MLLMs to VideoQA through question decomposition and dynamic token/frame compression, achieving a 29% relative accuracy gain on egocentric videos with LLaVA-NeXT-7B.

Adobe Research — San Jose, CA
Research Intern | *Mentors: Dr. Zhaowen Wang, Dr. Simon Jenni, Dr. Jing Shi* — May 2025 – Nov 2025

[ECCV'26] Compositional Layout Understanding in MLLMs

- Identified semantic drift and structural ambiguity as key challenges in compositional visual reasoning, and developed MASON with metadata-visual grounding and 3D structural perception, achieving 26% and 15% relative improvements over Gemini-2.5-Pro and GPT-5/GPT-o3, respectively, using Qwen2.5-VL-7B as the backbone.
- Constructed CoDeLayout, a ~20K-sample benchmark, via an automated pipeline for compositional structure mining and multimodal QA generation.

Academic Service

Conference Reviewing: ICLR; NeurIPS; ACL; EMNLP; WWW; IEEE Face & Gesture (FG)
Journal Reviewing: ACM TKDD

Skills

Languages: Python, LaTeX
ML Frameworks: PyTorch, Hugging Face Transformers, Hugging Face Datasets
Training/Fine-Tuning Libraries: Accelerate, PEFT, bitsandbytes
RL/Post-Training Frameworks: TRL, verl, OpenRLHF
Inference & Evaluation: vLLM, lmms-eval
Video Processing: ffmpeg, PyAV, decord
Systems/Infrastructure: Linux, Git, Slurm

Awards

National Scholarship, China	2021
Outstanding Student, Xidian University	2022
1st Prize, Undergraduate Computer Design Competition (National Level), China	2021
2nd Prize, RoboMaster National Robotics Competition, China	2019
3rd Prize, ICRA AI Challenge	2019